

IDMERS

Integrated Disaster Management and Early Warning System

Video Script | Runtime: ~2 minutes

0:00 – 0:10

OPENING HOOK

Visual: Monsoon rain, flooded streets, landslide aftermath footage/animation

Every monsoon season, Sri Lanka faces the same disaster twice — floods that swallow homes, and landslides that bury villages. Most of it, we could see coming. This is IDMERS — an integrated system built to predict, warn, and respond before disaster strikes.

0:10 – 0:35

COMPONENT 1: Flood Severity Assessment and Mapping

Visual: River gauges, DEM terrain model, flood extent map overlay

Our flood component analyzes rainfall, river water levels, elevation data, historical flood records, land use, population, and satellite imagery to understand exactly how severe a flood could become — and who it will affect. The system generates clear flood severity maps, giving authorities the insight they need to plan evacuations before water rises.

0:35 – 1:00

COMPONENT 2: Landslide Prediction

Visual: Hillside terrain, rainfall data animation, risk heatmap with SHAP breakdown

On the hillsides, the danger is silent — until it isn't. Our landslide component uses machine learning to analyze historical landslide occurrences, rainfall patterns, and soil type, calculating real-time risk across high-risk districts. An Explainable AI layer shows exactly which factor is driving each warning, and delivers it through an accessible map — high-contrast for elderly citizens, and in Sinhala and Tamil, so every village understands.

1:00 – 1:25

COMPONENT 3: Human Impact and Decision Support

Visual: Population density overlays, priority ranking dashboard

But knowing a hazard is severe isn't enough — we need to know who it hurts most. This component combines flood and landslide severity with population and vulnerability data, turning raw risk into a clear decision: which area needs help first, and what action to take — passed straight to the response

team.

1:25 – 1:50

COMPONENT 4: Emergency Response and Resource Coordination

Visual: Dispatch map, ambulance/rescue routes avoiding hazard zones

When every minute counts, this component turns priority alerts into action — automatically dispatching resources, prioritizing the hardest-hit districts during shortages, and generating safe evacuation routes that steer clear of active floods and landslides.

1:50 – 2:00

CLOSING

Visual: All four maps merging into one unified dashboard, then team names on screen

Four components. One connected system. From prediction to response — IDMERS is disaster management, reimagined for Sri Lanka.

TEAM IDMERS	
Flood Impact Assessment	Sinhalagoda S.A.U.R.A.V
Landslide Prediction	Bagya R.M.S.
Human Impact & Decision Support	Prathibhani Y.L.O.L.
Emergency Response & Coordination	Palihena M.B.

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